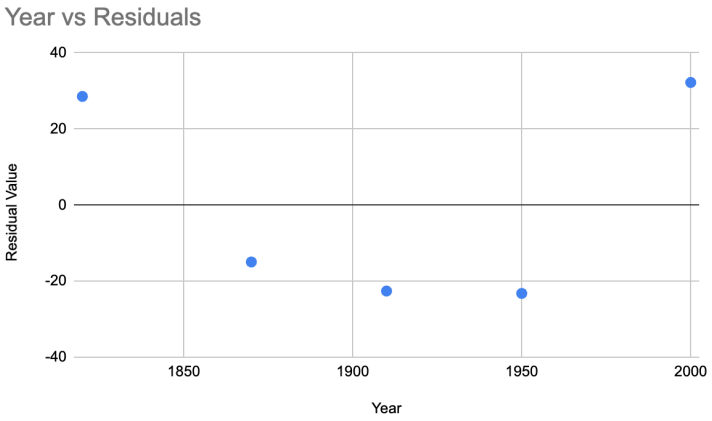
UID:
Due 2/10 at 2 PM

Data and Bias

1.
    a. There is voluntary bias, since people are volunteering to respond to the survey or not, rather than being required to. People with certain habits might be more inclined to answer the survey and thus volunteer to. There is under-coverage bias, since she makes conclusions about UCLA students but only asks the members of a student health and wellness club. This does not include all types of UCLA students, and could skew results since members in the group are probably likely to put more effort into wellness and sleeping habits. There could be response bias, since people were asked to self-report their sleeping schedule. People could have reasons for not telling the truth, like wanting to seem like they got more or less sleep than they actually did. This can result in unreliable data. The problem did not say if everyone in the club responded, but if they did not there could be non-response bias, since a specific type of people could not respond, skewing the results. There is also convenience bias, since the researcher went to a group of students that was convenient to ask rather than sampling from the population the right way.
    b.
        i. If there is bias in the training data, it will affect the model's predictions and thus introduce bias in the resulting decisions too. For example, if historically people of lower-income neighborhoods had a history of being denied loans due to discrimination, then the model learns from that relationship and thinks it has identified the correct pattern. Then, we will see discrimination against lower-income neighborhoods by the model's decisions too.
        ii. This might seem like it would remove specifically discrimination based on the location-related data, but there are probably other variables that are indirectly related to location and thus indicative of lower-income neighborhoods. For example, factors such as education, occupation, or race can be correlated with low-income neighborhoods and a factor in the location-based bias. Thus, the model would still have bias if those factors are there.

Linear Regression Calculations for 1 a-c

1. $B_1 = 1.491$; $B_0 = -2732.678$
    a. Population in 2020 = $B_0 + B_1(2020) = -2732.678 + (1.491*2020) = 279$ million
    b. R-squared value: 0.968. Since this value is close to 1, we can say it is better than the average value, so the estimated regression line fits the data well in the way that it is better than the naive model.



c. I do not think the linear assumption is correct, so this is not a good model. The relationship seems to be better suited by a quadratic function rather than a linear regression.

2. There is a somewhat strong correlation between heart disease and wine consumption since both the regular and adjusted R-squared values are around 70%, which means the model performs better than the average model. We cannot conclude that drinking more wine will reduce the risk of heart disease. The plot only shows a correlation between the two, not a causal relationship. For example, we can say that there is a correlation between wine consumption and heart disease, but not that just because someone consumes more wine they will be less likely to die of heart disease.
3.
    a.

	B_0 (intercept)	B_1 (slope)
Business Growth based on Investment	1.101	0.622
Investment based on Profit	40.000	7.625

    b.

	R-sq
Business Growth based on Investment	0.589
Investment based on Profit	0.605

Based on these R-squared values, business growth is moderately correlated with investment, this makes sense since if you are investing money in your business like buying high quality products or spending on marketing, you will have more success in your business, leading to business growth. Investment is also moderately correlated with profit. If you are profiting more money, then you have more money available to invest, so this correlation would make sense. Also, if you are investing more in your business, as explained above, you bring in more business and thus profit.

4.
    a. B_0: 5.541; B_1: 0.098

b. The r-squared value was 0.2, so there is not a strong linear relationship between X and Y.
c. The p-value of B_1 is 0.000, so we reject the null hypothesis. This suggests that X has a statistically significant impact on y. (used statsmodels .summary())
d. $B_1 + 2SE(B_1) = 0.098 + 2(0.006) = 0.110$; $B_1 - 2SE(B_1) = 0.098 - 2(0.006) = 0.086$. (used statsmodels .summary())

This results in a confidence interval of 0.086 to 0.110. The 95% confidence interval does not suggest that B_1 is meaningfully different from 0 since it is not greater than or equal to 1.

e. My answers for c and d contradict each other in that c suggests statistical significance, but d does not. The reason for this contradiction could be that we are using different metrics to evaluate the model in each part. We can solve this issue by better evaluating/deciding on which metrics to use. We should always interpret metrics in context when analyzing data, so we can decide which is more relevant to our issue.

5.

a. R-sq = 0.75. Since the value is above 0.5, it is better than the mean value. Since it is less than 1, the model is not perfect. However, since the value is somewhat high it shows that the linear model is a good model in the sense that it makes better predictions than if we were to use the average value.
b. A 95% prediction interval based on this question would be (68.11, 95.81). This can be interpreted as the following: given a duration of 5 minutes for the current aftershock, we expect with 95% confidence that there are between 68.11 and 95.81 minutes till the next aftershock. (used scipy.stats for t value and used the formula)
c. The model predicts that you have 81.96 minutes until the next aftershock, if the last aftershock lasted 5 minutes. We also predicted that the next aftershock will be in the interval above with 95% confidence. Based on these results, if you are leaving in 50 minutes, which is under the lower end of the interval, it is unlikely (but not definitively a no) that you will experience the next aftershock before leaving.

Interpretation of Coefficients in Linear Regression

a. Type 2 of encoding makes the most sense for A and 1 for B. This is because there is no ordering of the fish in dataset A. However, with B there is some comparison to be made with a hierarchy–for example, sashimi-grade salmon is better than both commercial and canned salmon, while commercial salmon is seen as better than canned salmon. Therefore, assigning them labels of 1, 2, and 3 makes sense.

b.

Dataset A: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2^{Tuna} + \beta_3 X_2^{Swordfish} + \beta_4 X_2^{Blobfish} + \beta_5 (X_1 X_2^{Tuna}) + \beta_6 (X_1 X_2^{Swordfish}) + \beta_7 (X_1 X_2^{Blobfish}) + \varepsilon$

Dataset B: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon$

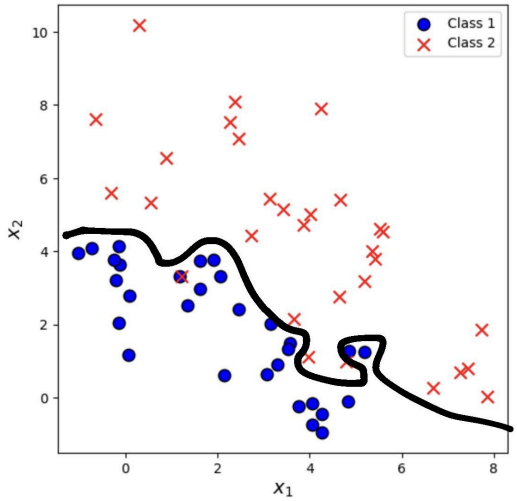
c.

β_0: Sales when the weight is 0 and the fish is not swordfish, blobfish, or tuna.
β_1: For every one unit increase in weight of fish, sales will increase by β_1.
β_2: Additional change in sales from baseline when fish is tuna.

β_3: Additional change in sales when fish is a swordfish.
β_4: Additional change in sales when fish is a blobfish.
β_5: Additional change in sales for an increase in weight when the fish is tuna.
β_6: Additional change in sales for an increase in weight when the fish is swordfish.
β_7: Additional change in sales for an increase in weight when the fish is blobfish.

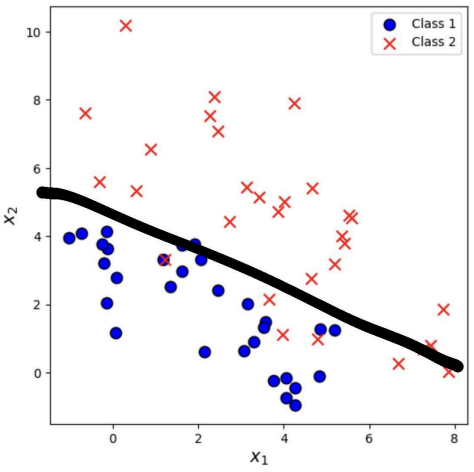
Bias

a.



Very jagged and deep curves separating the blue points from red. If we are overfitting, the training error will decrease over time while the test error will increase. This is because the model is essentially memorizing the training data so it seems to fit very well on it (low error). However, it is not learning true trends in the data and thus not good at generalizing, hence why the test error would increase. Some potential reasons for overfitting to occur are that there are too many parameters, too many interaction terms, collinearity, or the degree of the polynomial is too high. I think the model will perform poorly on the test set since it is unseen data and the model is overfit and memorized the training data very well rather than finding a pattern.

b.



We would see a very loose curve across the graph, something like just a smooth line down the middle, rather than carving around all the points like you would see in overfitting. In underfitting, the training error will be high and the testing error will be high as well. Though there may be

some fluctuations, both will remain high while the model is underfitting. This is because the model is too simple and not learning true trends in the data and thus not good at labeling data. Some potential reasons for underfitting to occur are that the degree of the polynomial is too low or there are not enough parameters. I think the model will perform poorly on the test set since it is unseen data and the model is extremely simple and does not really know how to make the classifications.

c. The optimal solution for both images is where the shaded region touches the circular lines. The L1 regularization is the graph on the left, while L2 regularization is the graph on the right. This is because the left graph uses a diamond, and we see the betas being 0 (since the intersection is at the y-axis). On the other hand, the right graph uses a circle and is not quite 0, so it is ridge regularization. Lasso is good for when you want sparse features and need to pick some variables. Ridge is good for easy interpretation and when you want to reduce model complexity, but not get rid of features.

Logistic Regression

a. $1 / [1+ (e^0)] = ½$; Log odds: 0; Odds: 1; Probability: 50% or 0.5
b.

	Log Odds (original 0)	Odds (original 1)
Inc X_1 by 2	10	e^{10}
Dec X_2 by 1	2	e^2

c. Increasing the coefficients means we are increasing the log-odds additively by the increased value for each unit increase in X_i. Decreasing the coefficients means the new log-odds would be less by that factor for each unit of X_i. We also must consider the signs of the coefficients and input features. If the signs are the same and we increase the coefficient, then we will see that increase. But if the signs are different, increasing the coefficient would lead to a net decrease. Similarly, if the signs are the same and we decrease the coefficient, we see a decrease in the log odds. If the signs are different, decreasing the coefficient would lead to a net increase. If we increase or decrease the betas by a value, the odds will multiplicatively increase or decrease by a factor of e^{value} (depends on sign again).
d. The decision boundary would be $2 + 5X_1 - 2X_2 - 3X_3 = 0$. Points that satisfy this equation would be on it.
e. This could be because of multicollinearity. When we used X_1 and X_2 separately, we saw different coefficients than when used together. This indicates that they are highly correlated. This is problematic because we might not be capturing the true effects of the coefficients in the independent models, since there is a relationship between the two variables that we are not considering – leading to a worse model that cannot generalize as well on new data.